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Laplace's Demon and Not-So-Simple Harmonic Motion

1.1 Taming the Demon with Algebra

We begin by converting the 2nd order differential equation of $F = m\ddot{x}$ into *two, simultaneous first-order* differential equations:

$$\dot{x} = v \,, \tag{1}$$

and

$$\dot{v} = \frac{F}{m} \,. \tag{2}$$

The force F is usually known to you depending on your location $F \equiv F(x)$.

We now use central difference numerical schemes to approximate the derivatives $\dot{x}$ and $\dot{v}$,

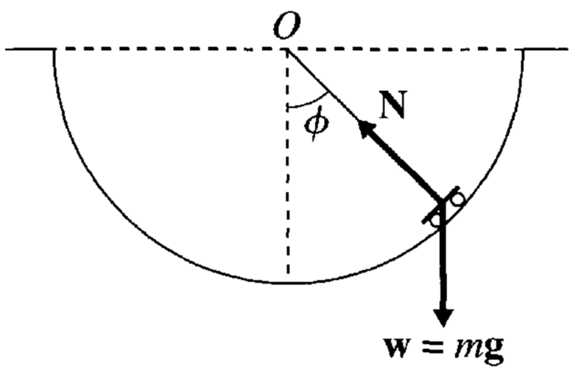
$$x(t + \Delta t) = x(t - \Delta t) + \big(2\Delta t\big)v(t) \tag{3}$$

and

$$v(t + \Delta t) = v(t - \Delta t) + \big(2\Delta t\big)\left(\frac{F(x(t))}{m}\right) . \tag{4}$$

Following this, if we take our initial conditions to be the same for a first two time steps (a standard bootstrapping trick for such central difference numerical schemes), the algorithm can proceed recursively and iteratively.

1.2 Laplace's Demon at the Skatepark



Laplace's Demon on the skatepark at IIT Delhi as shown in the figure has the following equation of motion,

$$\ddot{\phi} \;=\; -\frac{g}{l}\sin\phi \,, \tag{5}$$

where l is the radius of the skate track (we called this R in the tutorial), and g is the acceleration due to gravity. This can be seen by decomposing the forces into radial and tangential directions. Radially, there is no acceleration (the particle is constraint on the skatepark circular track), so normal force N balances the component of gravity $mg\cos\phi$ in that direction. Tangentially, gravity's component $mg\sin\phi$ remains unbalanced and provides the force for acceleration.

To solve it using the central difference scheme as discussed, you can use this **Google Sheets** solution we have implemented `https://tinyurl.com/laplacedemonsolution`.

Finally, let's get the time period of oscillation for this full, non-linear oscillator to leading order dependence on its amplitude. In general we do expect dependence on the amplitude ϕ_0, let's call this dependence as $T(\phi_0)$.

For the case of **small-angle approximation**, we have the equation of motion to be,

$$\ddot{\phi} \approx -\frac{g}{l}\phi \,, \tag{6}$$

for which the "standard" time period τ **does not** depend on the amplitude ϕ_0,

$$\tau = 2\pi\sqrt{\frac{l}{g}} \,. \tag{7}$$

To understand the full non-linear oscillator, we looked at its energy in class,

$$E = K + U = \frac{1}{2}ml^2\dot{\phi}^2 + U(\phi) \,, \tag{8}$$

where the potential energy $U(\phi)$ is

$$U(\phi) = mgl(1 - \cos\phi) \,. \tag{9}$$

Since energy is conserved, we can use it to connect with the amplitude where the particle has no speed (hence no kinetic energy),

$$\frac{1}{2}ml^2\dot{\phi}^2 + mgl(1-\cos\phi) \; = \; mgl(1-\cos\phi_0) \,. \tag{10}$$

This allows us to connect the (angular) speed with the position of the particle,

$$\dot{\phi} \; = \; \pm\sqrt{\frac{2g}{l}}\sqrt{\cos\phi - \cos\phi_0} \,. \tag{11}$$

We then used the fact that,

$$\dot{\phi} = \frac{d\phi}{dt} \implies dt = \frac{d\phi}{\dot{\phi}} \,. \tag{12}$$

Since we know $\dot{\phi}$ as a function of ϕ from Eq. (11), we can integrate this equation to obtain the time period. We choose to go from $t = 0$ when $\phi = 0$ to $t = T/4$ when the particle is at the amplitude $\phi = \phi_0$, that is, a quarter of a period. This is what allows us to pick the +ve square root in Eq. (11) to give,

$$\frac{T}{4} = \sqrt{\frac{l}{2g}}\int_0^{\phi_0} \frac{d\phi}{\sqrt{\cos\phi - \cos\phi_0}} \,, \tag{13}$$

or

$$T = 4\sqrt{\frac{l}{2g}}\int_0^{\phi_0} \frac{d\phi}{\sqrt{\cos\phi - \cos\phi_0}} \,. \tag{14}$$

We then did a Taylor series expansion for the cosine terms, but this time retaining to one higher order than we did for the simple harmonic case,

$$\cos\alpha = 1 \; - \underbrace{\frac{\alpha^2}{2}}_{\text{Simple Harmonic}} + \underbrace{\frac{\alpha^4}{24}}_{\text{First correction}} + \; \cdots \,. \tag{15}$$

Applying this for $\cos\phi - \cos\phi_0$ and got,

$$\cos\phi - \cos\phi_0 \approx \frac{(\phi_0^2 - \phi^2)}{2}\left[1 - \frac{(\phi_0^2 + \phi^2)}{12}\right] \,. \tag{16}$$

We can now invert under the square root, we get the following for the integrand of Eq. (14),

$$\frac{1}{\sqrt{\cos\phi - \cos\phi_0}} \approx \sqrt{\frac{2}{\phi_0^2 - \phi^2}}\left[1 - \frac{(\phi_0^2 + \phi^2)}{12}\right]^{-\frac{1}{2}} \tag{17}$$

We can now use a Binomial approximation $(1+\epsilon)^n \approx 1 + n\epsilon$ to simplify further,

$$\frac{1}{\sqrt{\cos\phi - \cos\phi_0}} \approx \sqrt{\frac{2}{\phi_0^2 - \phi^2}}\left[1 + \frac{(\phi_0^2 + \phi^2)}{24}\right] \tag{18}$$

Now, use ChatGPT or Gemini or something similar (use it since you are being given express permission to do so, the remainder of the parts should not use AI) to evaluate the integral of Eq. (14) with the approximation of Eq. (18). You will get the following answer,

$$\int_0^{\phi_0} \sqrt{\frac{2}{\phi_0^2 - \phi^2}}\left[1 + \frac{(\phi_0^2 + \phi^2)}{24}\right] d\phi \;=\; \left(\frac{\pi\sqrt{2}}{32}\right)\left(16 + \phi_0^2\right) \;=\; \left(\frac{\pi}{\sqrt{2}}\right)\left(1 + \frac{\phi_0^2}{16}\right). \tag{19}$$

Combine this back in Eq. (14), we get the time period of the non-linear pendulum to leading order depends quadratically on the amplitude,

$$T(\phi_0) \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\phi_0^2}{16}\right) . \tag{20}$$

1.3 More Interesting Things to Explore...Time Permitting

The solution for the 2D projectile motion under Earth's gravity is available in one of the tabs in the Laplace Demon Google Sheet: `https://tinyurl.com/laplacedemonsolution`.
